VIP Integrated Photonics

Course Information

VIP -001 (lecture) CRN 39690, 39688, 39692, 39691

Time: TBD

Location: TBD

Credits: Variable

Modality: Laboratory

Instructor

Office hours: TBD

If you cannot make these I will do my best to accommodate other times. Please send me an email with the subject line: **VIP OH** to request another time.

During office hours I am happy to give guidance on current homeworks and to discuss the details of old problems. I will also, of course, address any other questions you have or talk about any topics of interest.

Discussion board: To aid in discussion of the course topics, homework, and the like we will be "Conversations and Questions" discussion board on Brightspace. I encourage you to post questions here as opposed to emailing them. Hopefully, this will provide an efficient way to get questions answered quickly from classmates and myself. While I will participate in posting responses, I prefer student-student discussion.

Course Description

Analysis, modeling, and simulation of various integrated photonic components and devices.

Course Prerequisites and Co-requisites: This class is designed for advanced undergraduate students. In terms of background material, you should be comfortable with these topics: rudimentary differential equations, matrix algebra, basics of electromagnetism (Maxwell's equations), and the propagation, reflection, and refraction of plane waves, basics of Fourier series and Fourier transforms.

Learning Resources, Technology & Text

All class materials will be posted on Brightspace. In addition to homework, there may also be additional notes. Below is the link to the class site:

Brightspace:

To aid in discussion of the course topics, homeworks, and the like we will be using the "Conversations and Questions" discussion board on Brightspace. I encourage you to post questions here as opposed to emailing them. Hopefully, this will provide an efficient way to get questions answered quickly from classmates and myself. While I will participate in posting responses, I prefer student-student discussion.

Discussion board: "Conversations and questions" on Brightspace

- **Required Text:** *Silicon Photonics Design*, Lukas Chrostowski and Michael Hochberg, Publisher: Cambridge University Press, 1st edition (2015); Language: English; ISBN 978-1-107-08545-9

  Available online through through the Purdue Libraries

- **Additional reading:** Additional notes may be posted at times through the semester. You are expected to read these in addition to the course text.

- Your Brightspace landing page has a variety of links for academic success and policies, the Purdue Libraries and online writing lab, as well as the Disability Resource Center.

- **Use of artificial intelligence (AI) tools in this course:** The assignments used to assess your performance in this course are expected to reflect your own work. With that in mind, AI tools (e.g. ChatGPT, etc.) are not to be used on graded assessments.

Course Learning Outcomes:

A student who successfully fulfills the course requirements will demonstrate a knowledge if integrated photonic components and their design.

ASSIGNMENTS AND GRADING

Each student's course grade will be based upon the total numerical score he or she earns, with the following point allocation for homework, research poster, project, and engagement:

Table 1: Table of general course assignment categories and weighting.

Assessments	**Points**
Homework 1	10
Homework 2	10
Homework 3	10
Research poster	10
Project	40
Engagement	20
TOTAL	**100**

Homework assignments will be distributed and collected via Brightspace. Check Brightspace for any updates to the assignments, syllabus, or grading scale.

HOMEWORK

Doing the homework is the only way to truly learn the subject matter. Students who do not work the homework themselves typically do not perform well in the course.

- Due on the date and time specified on Brightspace. Submissions will be electronic via upload to Brightspace. Any changes in due date will be announced in class and / or on Brightspace.

- Please write your solutions legibly and in an organized manner and, where possible, place your final answer in a box. This will help the grader (me) follow your solution. Please upload good-quality solution files. For iPhone users, the Notes app will let you scan to pdf. For Android users, CamScanner comes recommended but I cannot personally attest to its performance.

  Illegible or poorly-organized homework may result in reduced or zero score for the assignment.

- You may work together as you solve your homework problems, as this can be an effective means of learning the material. If you do work in a group, please be sure that the solution you turn in is your own work. You will receive *reduced or zero credit* for homework submissions that appear to be copies of each other.

- Solutions to the homework assignments will be posted online shortly after they are due. Please review them and take advantage of office hours if they are unclear!

Research poster

Each student will contribute to a research poster for the Spring Undergraduate Research Conference. The poster will be graded based on completion - you must contribute to receive these points!

Presentations

Each student will give a presentation on a topic of their choice (relevant to the course material).

Topics: You will choose the topic - Prof. McKinney and the TAs are available to discuss.

Presentation details: Each student will present their topic in a ~15 minute talk. Presentations will he held during the last week of class.

Engagement

These points will be awarded based on the student's engagement during the class. Assignment completion, quantity and quality of participation in questions and discussion (e.g., in class, Brightspace, office hours, ect) will factor into this assessment. *Asking questions, getting help, and helping others is strongly encouraged!*

Grading Scale

In the past, I have used the following grade scale: A: 85-100, B: 70-84, C: 55- 69, D: 40-54. However, this scale may vary based on my perception of student performance. Plus/minus grading will be used, generally for students whose score falls near a boundary.

Class Attendance

Your attendance in class is important. If you must miss a class, you are responsible for any material, information, handouts, announcements, etc. that you missed. Since participation is a key part of your grade, repeated absences will detract from your overall grade. Please see the next section for absences related to illness.

Academic Guidance in the Event of Quarantine / Isolation or Emergent Illness

If you must miss class at any point in time during the semester due to illness, please reach out to me via Purdue email so that we can communicate about how you can maintain your academic progress.

For other emergent health issues, please see the University policy on medically excused absences. Should such a situation arise, please notify your advisor and me so that we can make arrangements.

Instructor option / class participation points will be adjusted on a case-by-case basis in accordance with the University regulations.

Academic Integrity

The ECE faculty expects every member of the Purdue community to practice honorable and ethical behavior both inside and outside the classroom. Academic integrity is a key value at Purdue University, as summarized by Purdue's Honor Pledge: > *"As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue."*

Any actions that might unfairly improve a student's score on homework, quizzes, or examinations will be considered cheating and will not be tolerated. Examples of cheating include (but are not limited to):

- Requesting a regrade of answers or work that has been altered.
- Submitting homework that is not your own or engaging in forbidden homework collaborations (including websites such as Chegg, Stack Exchange, ChatGPT, etc.).
- Plagiarism

At the instructor's discretion, cheating on an assignment will result in a reduced score, a zero score, or a failing grade for the course. All occurrences of academic dishonesty will be reported to the Assistant Dean of Students and copied to the ECE Associate Head for Education. If there is any question as to whether a given action might be construed as cheating, please see the instructor before you engage in any such action. Students may also report issues relating to academic integrity through the open OSSR by either emailing integrity@purdue.edu or by calling 765-494-8778. Such reporting may be done anonymously.

Use of copyrighted materials: Students shall not make course notes or materials available for others. The University Regulations: 'Always assume the materials presented by an instructor are protected by copyright unless the instructor has stated otherwise. Students enrolled in, and authorized visitors to, Purdue University courses are permitted to take notes, which they may use for individual/group study or for other non-commercial purposes reasonably arising from enrollment in the course or the University generally'.

In general, using language suggested by the University, any notes or other materials posted by me should be "considered to be 'derivative works' of my presentations and materials, and they are thus subject to my copyright in such presentations and materials." So, for example, you may not post any homeworks, exams, solutions, slides, or notes provided by me to any online platforms for any purpose.

Use of Artificial Intelligence (AI) Tools

The assignments used to assess your performance in this course are expected to reflect your own work. With that in mind, AI tools (e.g. ChatGPT, etc.) are not to be used on graded assessments.

Nondiscrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life.

Purdue's Nondiscrimination Policy Statement

Accessibility

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247. More details are available on our course Brightspace under Accessibility Information.

Mental Health Statement

Although your undergraduate years should be a special time in your life, university life can be stressful for a variety of reasons. Sometimes such stresses lead to serious mental health concerns. The Office of the Provost has suggested the following statements to inform you of resources in case of need.

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online, a new web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS). TAO is available to students, faculty, and staff at any time.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 am- 5 pm.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office on the second floor of the Purdue University Student Health Center (PUSH) during business hours.

Emergency Preparation

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

A link to Purdue's Information on Emergency Preparedness is located on our Brightspace under "University Policies and Statements." This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System. I encourage you to download and review the Emergency Preparedness for Classrooms document (PDF) or (Word).

The first day of class, I will review the Emergency Preparedness plan for our specific classroom, following Purdue's required Emergency Preparedness Briefing. Please make note of items like:

- The location to where we will proceed after evacuating the building if we hear a fire alarm;
- The location of our Shelter in Place in the event of a tornado warning;
- The location of our Shelter in Place in the event of an active threat such as a shooting.

Basic Needs Security

If you are facing challenges securing basic needs such as food, housing, transportation, health services, or access to technology or childcare resources and believe this may affect your performance in the course, please contact the Office of the Dean of Students (ODOS) to help coordinate with community resources. These services vary by location. In West Lafayette, see the Basic Needs Program website, or email basicneeds@purdue.edu.

Course Evaluation

Toward the end of this semester, you will be provided with an opportunity to give feedback on this course and your instructor. Purdue uses an online course evaluation system, and I will not have access to this anonymous feedback until after final grades are submitted. You will receive an official email from evaluation administrators with a link to the online evaluation site and will receive a prompt to complete the survey when you login to Brightspace. The subject line will be: Please take 2-5 minutes to complete the survey. Check your "Junk E-mail" folder occasionally to be sure the evaluation emails were not accidentally routed there. Your participation is an integral part of this course, and your feedback is vital to improving education at Purdue University. I strongly urge you to participate in the evaluation system.